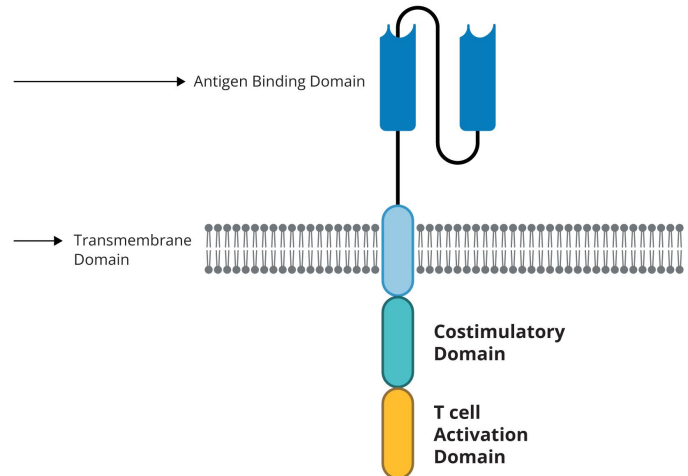


A "living drug"

HOW CAR T-CELL THERAPY WORKS

CAR T-cell therapy uses a patient's own T cells as a **"living drug."** The process begins by collecting and separating these T cells from the patient's blood.

Next, the cells are **genetically engineered** in a laboratory to produce **chimeric antigen receptors (CARs)**, allowing them to recognize specific cancer antigens. They are then expanded to hundreds of millions and infused back into the patient.



CAR STRUCTURE

Image: National Cancer Institute

The entire manufacturing process takes about **3 to 5 weeks**. Once infused, the CAR T cells continue to multiply in the body and actively kill cancer cells bearing the **target antigen**.

This approach represents a significant advancement in **personalized medicine**, offering new hope for patients with certain types of **blood cancers**. Researchers are continuously working to improve its safety and expand its application to other diseases.